

Abhay Mahajan

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AI/ML undergraduate and Generative AI Intern with hands-on experience building LLM-powered applications, RAG systems, agentic workflows, and full-stack AI products. Currently interning at Corover.ai shipping conversational AI systems. Built Clarix, an AI study platform with RAG-based Q&A, adaptive quizzing, and gamification, using Next.js, FastAPI, PostgreSQL, and ChromaDB. Seeking a full-time GenAI / LLM Engineer role.

EXPERIENCE

Corover.ai | Generative AI Intern

Jan 2026 – Present

- Built LLM-powered conversational AI systems handling multi-turn dialogue with tool-use and memory, deployed to production users
- Designed and shipped RAG pipelines for document-grounded Q&A, combining FAISS/vector retrieval with LangChain orchestration to reduce hallucinations vs. vanilla GPT responses
- Engineered agentic workflows using plan-execute patterns, enabling multi-step task decomposition across external APIs and knowledge bases

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning

Aug 2022 – Jun 2026

Symbiosis Institute of Technology, SIU, Pune

PROJECTS

Clarix — AI Study Platform (Founder / Solo Engineer) [GitHub ↗](#)

Feb 2026 – Present

- Architected a full-stack EdTech SaaS with Next.js frontend, FastAPI backend, PostgreSQL, and ChromaDB — covering auth, RAG pipeline, quiz engine, gamification, and admin panel
- Built a RAG-based AI tutor that indexes user-uploaded PDFs into ChromaDB and generates contextually grounded explanations, reducing off-topic LLM responses
- Implemented adaptive quiz generation using LLMs with structured weakness detection: automatically identifies knowledge gaps and feeds a personalized revision planner
- Shipped production-grade features: JWT auth with HTTP Only cookies, Redis caching, brute-force lockout, email verification, referral system with XP rewards, and a Plausible analytics integration
- Designed CI/CD pipeline with GitHub Actions, 30+ backend tests, and Locust load testing; containerized with Docker Compose for VPS deployment

Agentic ChatBot [GitHub ↗](#)

Feb 2026 – Present

- Built a LangChain-based agentic system with tool-use, memory modules, and plan-execute workflow for multi-step reasoning over external APIs
- Integrated dynamic function-calling architecture allowing the agent to route tasks to domain-specific tools at runtime

LangChain RAG ChatBot

Jan 2026 – Feb 2026

- Built a local-first RAG chatbot using LangChain + Ollama (LLaMA 3) with FAISS vector search over custom document corpora; FastAPI backend for query serving

AeroVis — Drone Surveillance Model

Oct 2024 – Nov 2024

- Developed a drone-based monitoring model for real-time traffic and disaster management, providing aerial surveillance insights via a computer vision pipeline

TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript

GenAI / LLM: LangChain, Ollama, OpenAI API, RAG pipelines, Agentic workflows, Prompt engineering

Backend: FastAPI, PostgreSQL, Redis, ChromaDB, FAISS, Docker, GitHub Actions

Frontend: Next.js, React, Tailwind CSS

ML / Data: TensorFlow, scikit-learn, Pandas, NumPy

Tools: Figma, Adobe XD, Postman, Locust